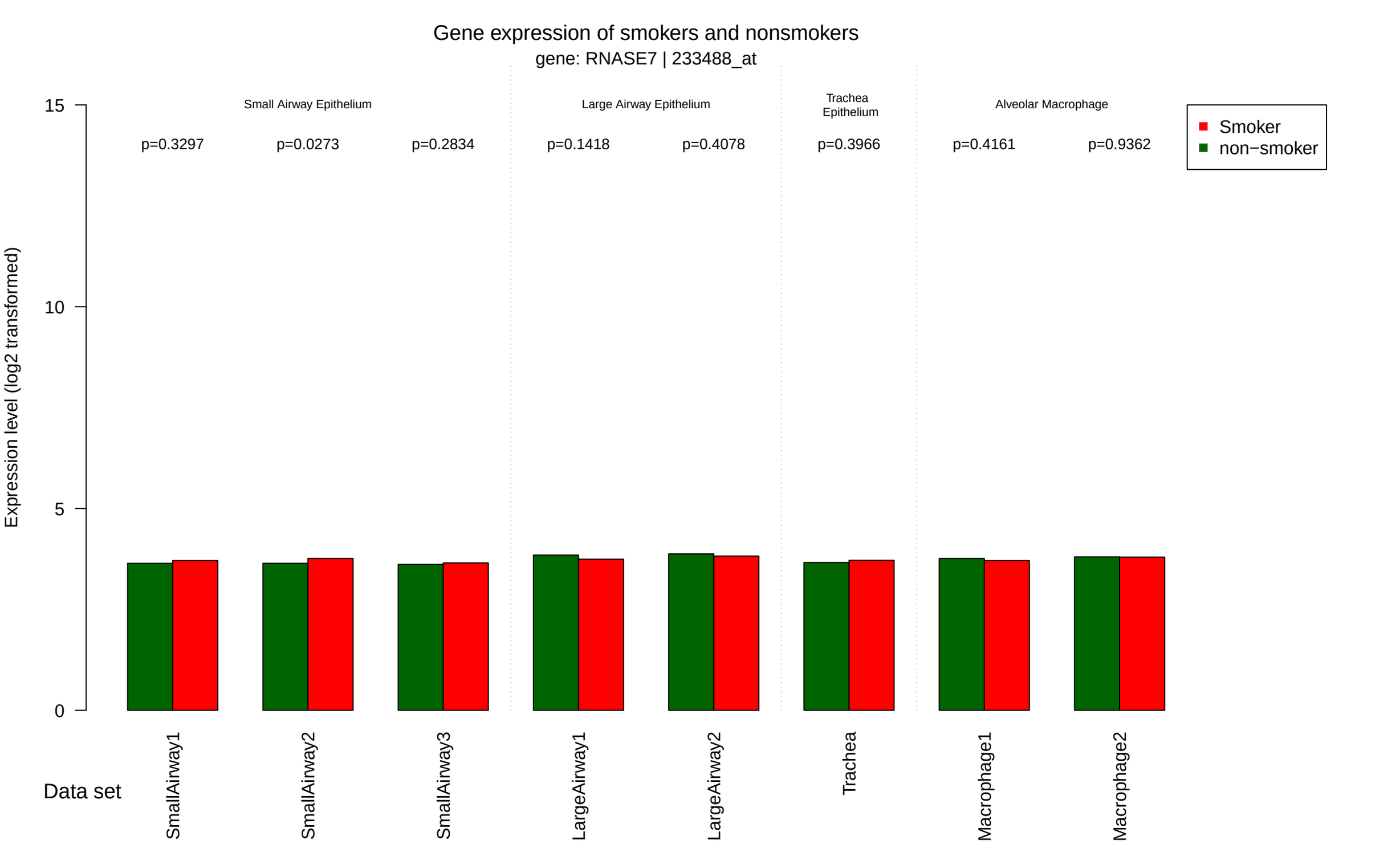
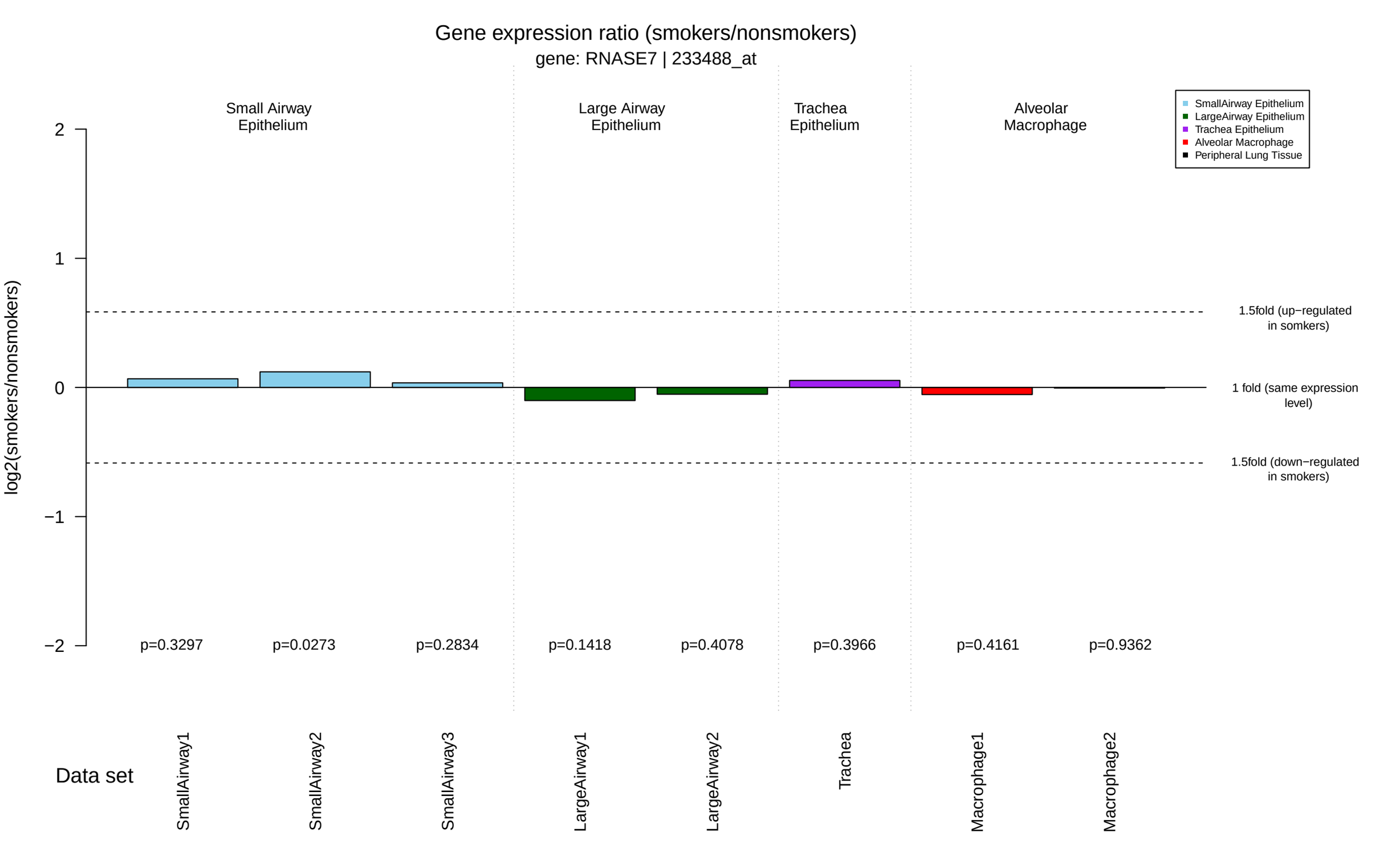
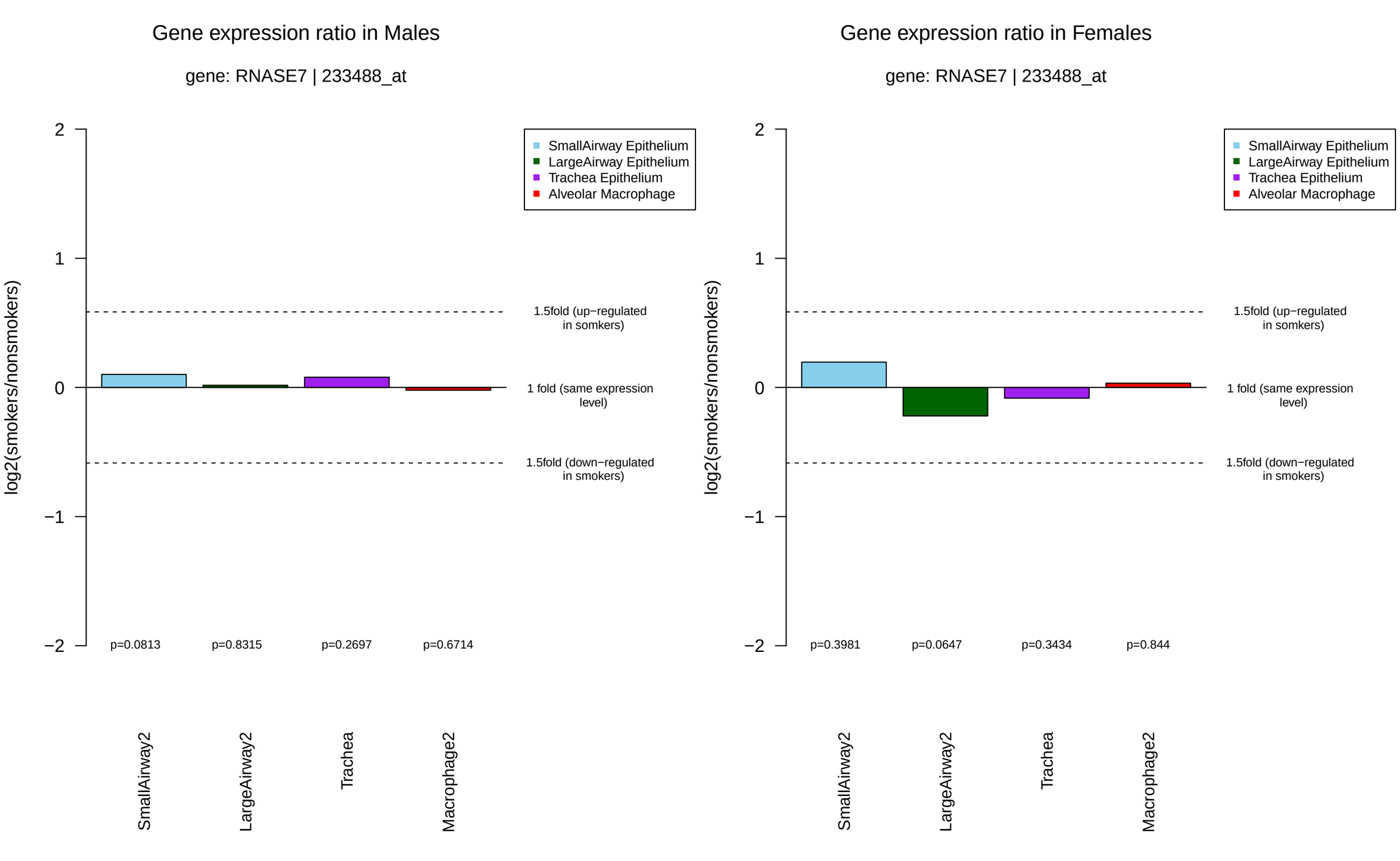
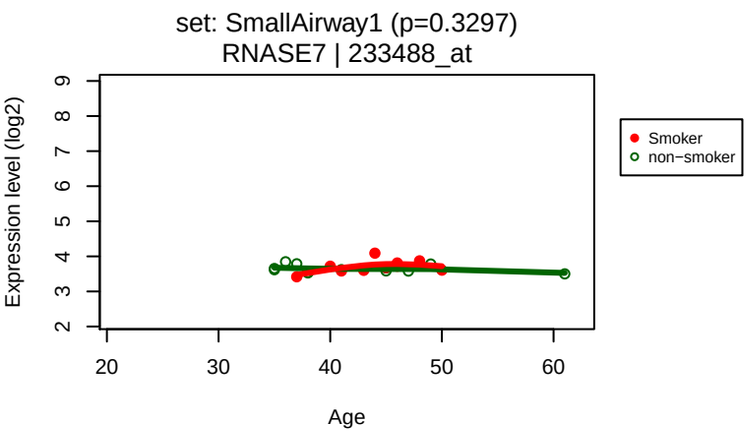
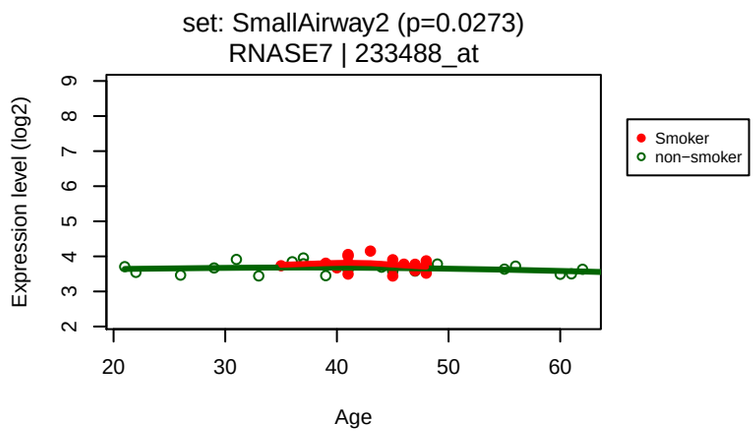
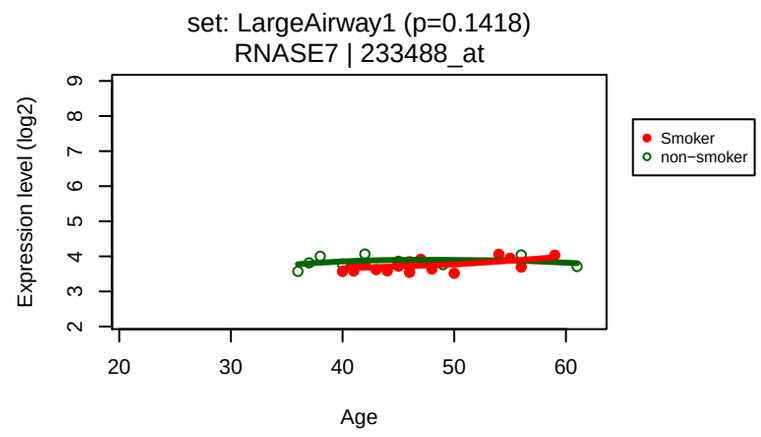

Gene expression of smokers and nonsmokers
gene: RNASE7 | 233488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3297
p=0.0273
p=0.2834
p=0.1418
p=0.4078
p=0.3966
p=0.4161
p=0.9362
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

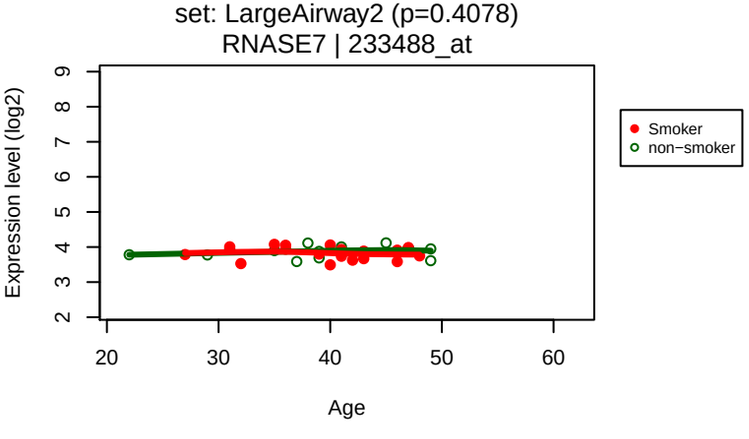

Gene expression ratio (smokers/nonsmokers)
gene: RNASE7 | 233488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3297
p=0.0273
p=0.2834
p=0.1418
p=0.4078
p=0.3966
p=0.4161
p=0.9362
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

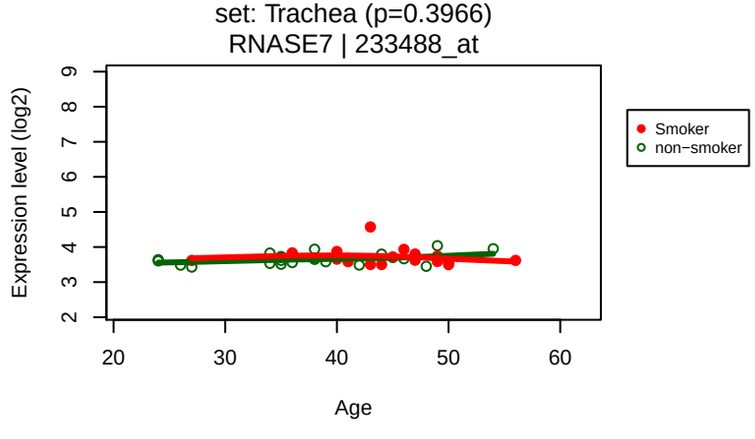

Gene expression ratio in Males
gene: RNASE7 | 233488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0813
p=0.8315
p=0.2697
p=0.6714
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNASE7 | 233488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3981
p=0.0647
p=0.3434
p=0.844
SmallAirway2
LargeAirway2
Trachea
Macrophage2

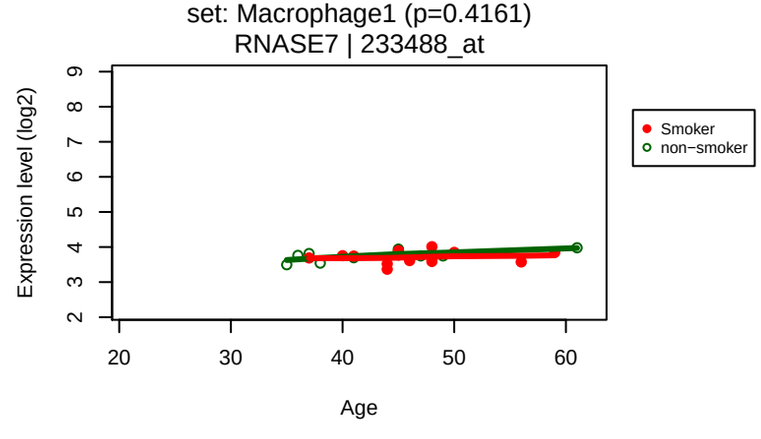

set: SmallAirway1 (p=0.3297)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker

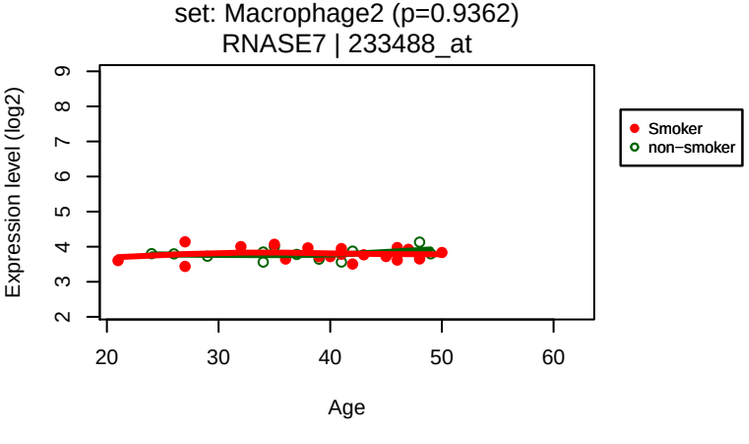

set: SmallAirway2 (p=0.0273)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker

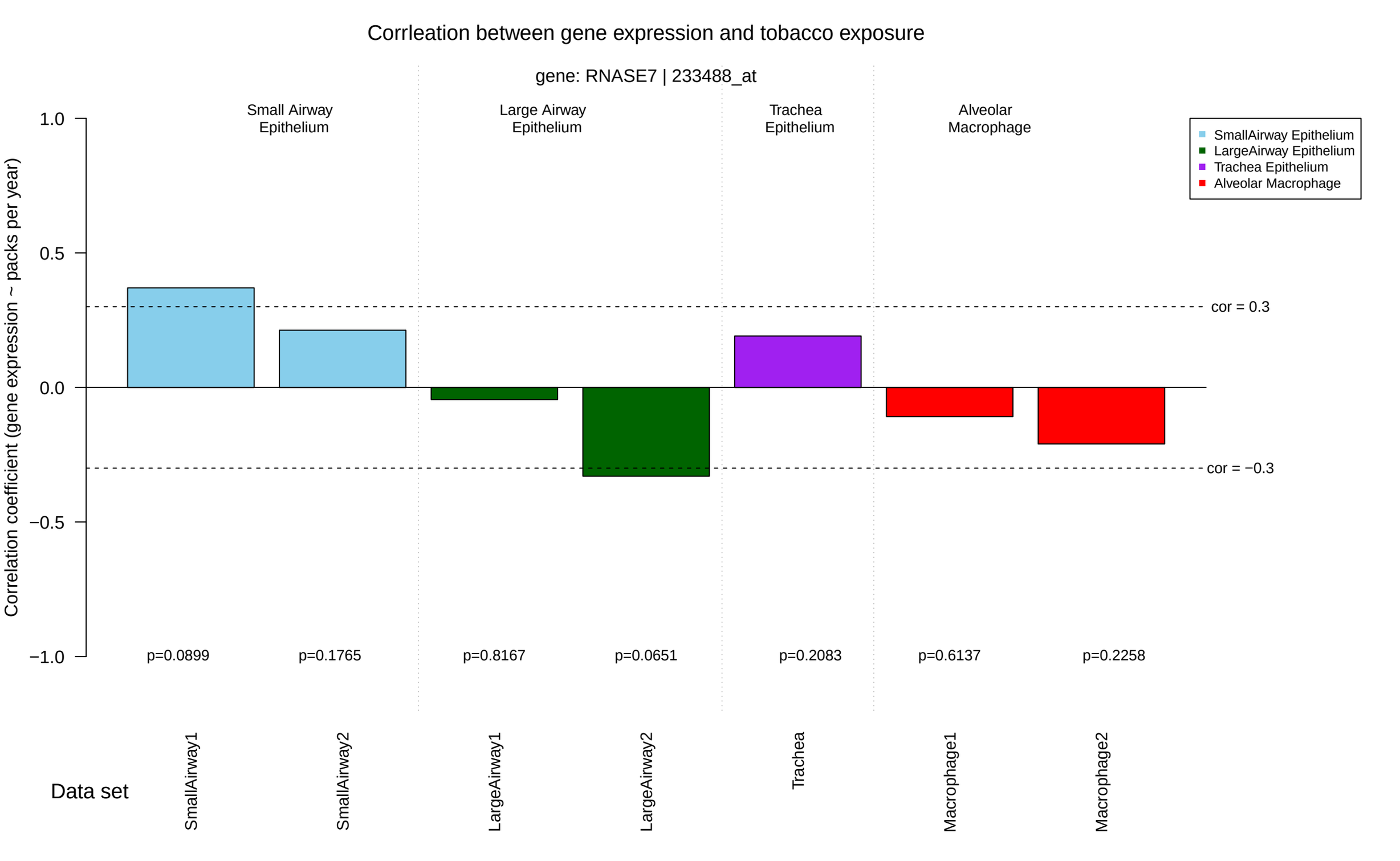

set: LargeAirway1 (p=0.1418)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4078)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3966)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4161)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9362)
RNASE7 | 233488_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNASE7 | 233488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0899
p=0.1765
p=0.8167
p=0.0651
p=0.2083
p=0.6137
p=0.2258
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2